

A Consensus-Based Practical and Daily Guide for the Treatment of Acne Patients

Abstract

Background

Many current guidelines provide detailed evidence-based recommendations for acne treatment.

Objective

To create consensus-based, simple, easy-to-use algorithms for clinical acne treatment in daily office-based practice and to provide checklists to assist in determining why a patient may not have responded to treatment and what action to take.

Methods

Existing treatment guidelines and consensus papers were reviewed. The information was extracted and simplified according to daily clinical practice needs using a consensus-based approach and based on the authors' clinical expertise.

Results

Separate simple algorithms are presented for the treatment of: - Predominant comedonal acne - Predominant papulopustular acne - Nodular/conglobate acne

- **Comedonal acne:** Initial treatment with topical retinoid, azelaic acid, or salicylic acid.
- **Papulopustular acne:** Fixed-combination topicals recommended, with therapy tailored to disease severity.

- **Nodular/conglobate acne:** Oral isotretinoin or fixed combinations plus oral antibiotics (in men); in women, these may be supplemented with oral anti-androgenic hormonal therapy.

Treatment response should be evaluated: - **8 weeks** after initiation for comedonal or papulopustular acne - **12 weeks** after initiation for nodular/conglobate acne

Once clear or almost clear, **maintenance therapy** with a topical retinoid or azelaic acid is recommended to prevent recurrence.

Reasons for lack of treatment response fall into five categories: 1. Disease progression

2. Non-drug-related reasons

3. Drug-related reasons

4. Poor adherence

5. Adverse events

Conclusion

This practical guide provides dermatologists with simple, clinically applicable treatment algorithms tailored to different acne types. The included checklists for evaluating treatment failure facilitate effective management in daily practice.

Introduction

Acne is a chronic inflammatory disease of the pilosebaceous unit, commonly affecting adolescents and imposing a significant psychosocial and physical burden. The primary pathogenic factors include: - Increased sebum production
- Abnormal keratinization
- Colonization by *Cutibacterium acnes* (formerly *Propionibacterium acnes*)
- Inflammatory response mediated by innate immunity

Due to this multifactorial pathophysiology, effective treatment typically requires combination therapy targeting multiple pathways.

Clinically, acne presents in various forms. This article focuses on three main types: 1. **Predominant comedonal facial acne**

2. **Predominant papulopustular facial acne**

3. **Nodular and/or conglobate acne** (affecting face, trunk, and upper limbs)

While numerous national and international guidelines offer evidence-based recommendations, there remains a need for simplified, practical tools that can be easily applied in routine clinical settings.

This article presents: - Consensus-based treatment algorithms for the three main acne types

- Checklists to identify causes of treatment failure
- Guidance on next steps when response is inadequate

Special populations such as infants, adult women with hormonal acne, and syndromic acne are not addressed here.

Methods

The treatment algorithms were developed by a panel of 17 acne experts. The process involved: - Review of key international, evidence-based acne guidelines, including those from: - Global Alliance to Improve Outcomes in Acne

- European Dermatology Forum
- American Academy of Dermatology
- Asian expert groups

The data from these sources were synthesized, simplified, and adapted to reflect real-world clinical practice through expert consensus and clinical experience. Recent literature was also considered to support recommendations.

Results

Treatment Algorithms

For ease of use in daily practice, separate algorithms are provided for:

1. Predominant Comedonal Acne

First-line treatment: - Topical retinoid

- Azelaic acid
- Salicylic acid

These agents normalize follicular keratinization and prevent comedone formation.

2. Predominant Papulopustular Acne

Recommended: - Fixed-combination topical therapies (e.g., benzoyl peroxide + antibiotic, retinoid + benzoyl peroxide, or retinoid + antibiotic)

- Therapy should be adjusted based on severity

Combination products improve efficacy and reduce antibiotic resistance risk.

3. Nodular/Conglobate Acne

Treatment options: - **Oral isotretinoin** – first-line for severe nodular or conglobate acne

- **Fixed combinations + oral antibiotics** – alternative in men if isotretinoin is contraindicated or declined
- **In women:** Consider adding **oral anti-androgenic hormonal therapy** (e.g., combined oral contraceptives, spironolactone) as adjunctive treatment

Evaluation of Treatment Response

Assessment should be performed using the **Global Evaluation Acne (GEA) scale**.

- **Clinical improvement:** Reduction of at least one to two grades on the GEA scale
- **Timing of evaluation:**
- **8 weeks** for comedonal and papulopustular acne
- **12 weeks** for nodular/conglobate acne

Maintenance Therapy

Once a patient is clear or almost clear: - Initiate **maintenance therapy** with:

- Topical retinoid
- Or azelaic acid
- Purpose: Prevent recurrence by maintaining follicular patency and reducing inflammation

Checklist: Causes of Treatment Failure

When a patient fails to respond, consider the following five categories:

1. **Disease progression**
 2. Natural course of acne may worsen before improving
 3. Consider escalation of therapy if progression occurs
4. **Non-drug-related reasons**
 5. Hormonal fluctuations (e.g., menstrual cycle, PCOS)
 6. Lifestyle factors (diet, stress, skincare practices)
 7. Cosmetic product use (comedogenic ingredients)
8. **Drug-related reasons**
 9. Inappropriate choice of agent for acne type/severity
 10. Suboptimal dosing or formulation
 11. Drug interactions
12. **Poor adherence**
 13. Common due to complex regimens, side effects, or lack of immediate results
 14. Simplify regimen, educate patient, and assess understanding
15. **Adverse events**
 16. Irritation, dryness, or allergic reactions may limit use
 17. Adjust formulation, frequency, or switch agents

For each identified cause, appropriate corrective actions should be taken, such as modifying treatment, improving patient education, or switching therapies.

Note: Figures and tables referenced (e.g., Fig. 1–4, Table 1) are not included in this text version.